LLM-Informed Bayesian Priors for Reducing Experimental Cost in Behavioral Economics

Final Report — CSE 8803 IUQ, Spring 2026

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Code: https://github.gatech.edu/bfeng66/cse8803-iuq-project

Video: https://mediaspace.gatech.edu/channel/Spring+2026+CSE+8803+IUQ[1]

Abstract

Behavioral economic experiments are expensive: each datum costs participant fees, lab time, and IRB overhead. We ask whether large language model (LLM) simulations can serve as an informative Bayesian prior, reducing the human sample size needed to reach a target posterior precision. We model dictator-game sharing ratios with a Beta likelihood and compare two pipelines: (i) a flat-prior baseline solved by Metropolis–Hastings MCMC, and (ii) a frontier method that encodes 500 GPT-4o pseudo-observations as a power prior [Chen and Ibrahim, 2000] and solves the posterior with Stein Variational Gradient Descent [Liu and Wang, 2016]. On 137 dictator observations from von Blanckenburg et al. [2023], the LLM prior achieves a sample-size reduction ratio $\rho = n^*_{\text{LLM}}/n^*_{\text{flat}} \approx 0.26$ on the giving game and $\rho \approx 0.55$ on the taking-game variant; coverage of 95% posterior-predictive intervals tracks nominal under all but moderate upward prior misspecification, where it falls to ∼83%. A leakage ablation with `gpt-3.5-turbo` (pre-Horton2023 cutoff) recovers an identical $\rho = 0.26$ but with markedly worse coverage (0.83 vs 0.98 at n=50), indicating that the gpt-4o advantage is calibration quality, not sample-size benefit, and that the result is not a pure training-data leakage artefact. We report the compute–accuracy trade-off, document the regimes where the LLM prior helps and hurts, and discuss why SVGD becomes the natural inference engine in higher-dimensional extensions of this template.

1 Introduction

Bayesian inference of human behavioral parameters is data-hungry. Eliciting tight posteriors on traits like generosity or risk aversion routinely requires hundreds of subjects per condition, at high marginal cost per observation. Yet a contemporary LLM, prompted with the same instructions an experimenter would read to a participant, can produce thousands of synthetic decisions in minutes [Horton et al., 2023, Argyle et al., 2023, Aher et al., 2023]. These "silicon subjects" are not human substitutes, but their distribution carries information about the behavior we are trying to estimate. The natural question for uncertainty quantification (UQ) is: *how much real data does an LLM prior save, and under what conditions does it harm calibration?*

Contribution. We instantiate a small but complete LLM-informed Bayesian pipeline on individual-level dictator-game data [von Blanckenburg et al., 2023], with the following ingredients: (1) a Beta likelihood on continuous sharing ratios; (2) a baseline flat prior solved by log-space Metropolis–Hastings (MH); (3) a frontier method with an LLM power prior solved by Stein Variational Gradient Descent (SVGD); (4) calibration via held-out coverage of 95% posterior-predictive intervals; and (5) a sample-size reduction ratio $\rho = n^*_{\text{LLM}}/n^*_{\text{flat}}$ that quantifies the experimental savings. On top of this baseline, we conduct four diagnostic studies that the project check-in did not address: a cross-game

[1] Channel URL; the direct video URL will replace this placeholder after upload to MediaSpace on Apr 27.

replication on the taking-game variant, a continuous misspecification curve over $\delta \in [-3, +3]$ prior std-deviations, a training-data leakage ablation against `gpt-3.5-turbo`, and a compute–accuracy Pareto analysis comparing wall-clock time and effective sample size of the two inference engines. The combination yields a fail-mode-aware accounting of when the LLM prior is worth using.

2 Related Work

LLMs as economic agents. Horton et al. [2023] introduces "homo silicus" as an LLM-driven simulation tool for economic experiments, replicating qualitative effects of classic studies. Argyle et al. [2023] and Aher et al. [2023] extend this to political and psychometric tasks. Recent working papers [Gao et al., 2026, Ludwig et al., 2024] formalize how to weight and combine LLM evidence with human data; the present work uses the power-prior construction [Chen and Ibrahim, 2000] as a transparent, single-knob instantiation of that idea.

Variational inference for posterior approximation. SVGD [Liu and Wang, 2016] approximates a posterior by a deterministic interacting particle system whose update direction is the kernelised score of the target. It scales to intermediate dimensions where MCMC mixing degrades but standard mean-field VI cannot capture posterior dependence. Our pipeline uses SVGD to demonstrate the design pattern; we honestly report below that for the two-parameter Beta posterior, MCMC is faster per equivalent ESS, and discuss why SVGD becomes attractive when the model dimension grows.

3 Method

3.1 Likelihood and quantity of interest

For each dictator-game observation $y_i \in (0, 1)$ (a sharing ratio), we adopt $y_i \mid \alpha, \beta \sim \mathrm{Beta}(\alpha, \beta)$ independently. With n human observations $\boldsymbol{y} = (y_1, \ldots, y_n)$ and prior $p(\alpha, \beta)$,

$$p(\alpha, \beta \mid \boldsymbol{y}) \;\propto\; \frac{1}{B(\alpha, \beta)^n} \prod_{i=1}^{n} y_i^{\alpha-1}(1 - y_i)^{\beta-1} \cdot p(\alpha, \beta). \tag{1}$$

Our primary scalar is the population mean sharing ratio $\mu = \alpha/(\alpha + \beta)$. The decision-relevant quantity is the *sample-size reduction ratio* $\rho = n^*_{\mathrm{LLM}}/n^*_{\mathrm{flat}}$, where n^* is the smallest n that brings $\mathrm{Var}(\mu \mid \boldsymbol{y}_{1:n})$ below a fixed target precision. We choose the target as the midpoint between the flat-prior posterior variances at the smallest and largest n in our sweep, so that ρ is well-defined and comparable across runs.

Sources of uncertainty. Four sources contribute, and the pipeline must account for each separately. (a) *Parameter uncertainty* in (α, β) – the inferential target, captured by the posterior in (1). (b) *Observation noise* – the irreducible Beta sampling spread of an individual y_i around the population mean, which dominates the posterior-predictive interval at large n even when parameter uncertainty is small. (c) *Model error* – the Beta family is the simplest parametric form for $(0, 1)$-valued data and may mis-fit multi-modal sharing distributions; we treat this as un-modeled and read it through coverage diagnostics. (d) *Prior input uncertainty* – the LLM persona that produces our pseudo-observations is itself stochastic across model versions, prompts, and temperature, and may drift from the human population in ways our experiment does not directly observe; the dense misspecification grid (§4.5) and cross-model leakage ablation (§4.6) probe this source.

3.2 Baseline: flat prior with log-space Metropolis–Hastings

We use $p(\alpha, \beta) \propto 1$ on $(0, 100)^2$ as an effectively flat prior. Both parameters are positive; we sample in $\boldsymbol{\eta} = (\log \alpha, \log \beta)$ with isotropic Gaussian proposals $\boldsymbol{\eta}' = \boldsymbol{\eta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, 0.3^2 I)$, and accept with the Hastings ratio

$$r = \min\left(1, \frac{\pi(\boldsymbol{\eta}')\, J(\boldsymbol{\eta}')}{\pi(\boldsymbol{\eta})\, J(\boldsymbol{\eta})}\right), \qquad J(\boldsymbol{\eta}) = \alpha\beta = e^{\eta_\alpha + \eta_\beta}, \tag{2}$$

i.e. multiplying by the log-transform Jacobian. We run 15,000 post-burn-in samples after a 3,000-step burn-in.

3.3 Frontier: LLM power prior with SVGD

We elicit $M = 500$ pseudo-observations $\tilde{y}_1, \ldots, \tilde{y}_M$ from GPT-4o (temperature 1.0, ten batches of 50, seed $42 + \text{batch}$) using the dictator-game prompt verbatim from the human protocol. Following Chen and Ibrahim [2000], we encode this LLM evidence as a *power prior*

$$p_{\text{LLM}}(\alpha, \beta) \;\propto\; \prod_{j=1}^{M} \text{Beta}(\tilde{y}_j; \alpha, \beta)^{w/M}, \tag{3}$$

with effective weight w controlling how many real observations the LLM is worth in posterior-information units. We use $w = 20$ throughout (sensitivity analysis below); this is roughly one-quarter of the held-out sample, large enough to matter but small enough not to drown the data.

For inference we run SVGD [Liu and Wang, 2016] with $K = 100$ particles $\{\boldsymbol{\eta}^{(k)}\}_{k=1}^{K}$ in log-space, RBF kernel $\kappa(\boldsymbol{\eta}, \boldsymbol{\eta}') = \exp\bigl(-\|\boldsymbol{\eta} - \boldsymbol{\eta}'\|^2/(2h)\bigr)$ with median-heuristic bandwidth, Adam step size 0.05, and $T = 2{,}000$ iterations. Each particle is updated by

$$\boldsymbol{\eta}_{t+1}^{(k)} \;=\; \boldsymbol{\eta}_t^{(k)} \;+\; \frac{\epsilon_t}{K} \sum_{\ell=1}^{K} \Bigl[\kappa(\boldsymbol{\eta}_t^{(\ell)}, \boldsymbol{\eta}_t^{(k)})\, \nabla_{\boldsymbol{\eta}} \log p(\boldsymbol{\eta}_t^{(\ell)} \mid \boldsymbol{y}) + \nabla_{\boldsymbol{\eta}_t^{(\ell)}} \kappa(\boldsymbol{\eta}_t^{(\ell)}, \boldsymbol{\eta}_t^{(k)})\Bigr]. \tag{4}$$

Particles are initialized at the method-of-moments estimate of $(\log \alpha, \log \beta)$ plus Gaussian jitter; gradients are computed analytically via $\psi(\cdot)$ derivatives of $\log B(\alpha, \beta)$, and clipped to $[-100, 100]$ for stability.

3.4 Calibration and diagnostics

We measure *empirical coverage*: for held-out observation y, we compute the posterior-predictive CDF $F(y) = \mathbb{E}_{(\alpha,\beta)\sim p(\cdot|\boldsymbol{y})}\, F_{\text{Beta}}(y; \alpha, \beta)$ using posterior samples and ask whether $F(y) \in [0.025, 0.975]$. Averaging over the held-out test set gives the empirical coverage of the nominal-95% interval. For MCMC we compute effective sample size for μ via the initial-positive-sequence estimator [Geyer, 1992]; for SVGD we use the number of particles as a construction-by-design ESS, while reporting wall-clock time.

4 Experiments

4.1 Setup

Data: 137 individual-level dictator decisions from von Blanckenburg et al. [2023], split as 69 *giving*-game and 68 *taking*-game observations. We held out 30% (41 obs) for calibration and used the

remaining 96 for inference. All sharing ratios are clipped to $[10^{-3}, 1 - 10^{-3}]$ to keep the Beta likelihood finite. All experiments use seed 42; LLM elicitation seeds are 42 + batch index per API call.

4.2 Sample-size reduction ρ on the giving game (replication)

Table 1 reports posterior variances of μ across $n \in \{5, 10, 20, 50, 90\}$ under flat and LLM priors with the gpt-4o pseudo-observations. The LLM prior reduces variance by a factor of 1.27× to 2.62× depending on n, with the largest gains at small n (where the prior carries the most weight in posterior information). Setting the precision target as the midpoint of the flat-prior endpoints yields $n^*_{\text{flat}} \approx 19.2$ and $n^*_{\text{LLM}} \approx 5.0$, i.e. $\rho \approx 0.26$ – the LLM prior recovers the same precision with a 74% smaller human sample. Figure 1 plots the variance–n curves on log-log axes.

n	$\text{Var}(\mu)_{\text{flat}} \times 10^3$	$\text{Var}(\mu)_{\text{LLM}} \times 10^3$	Ratio	Coverage 95% (LLM)
5	5.16	2.17	2.38×	0.98
10	4.81	1.84	2.62×	0.98
20	2.88	1.41	2.04×	0.98
50	1.15	0.81	1.42×	0.98
90	0.77	0.61	1.27×	0.98

Table 1: Giving game (gpt-4o LLM prior): variance reduction across sample sizes. The LLM prior helps most at small n; coverage tracks nominal at every n.

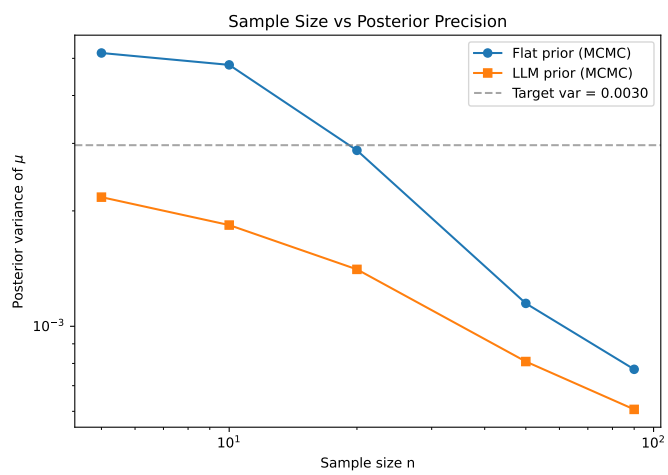

Figure 1: Posterior variance of μ vs sample size n, giving game. The LLM prior crosses the precision target near $n \approx 5$; the flat prior requires $n \approx 19$.

4.3 Cross-game replication (taking-game)

Phillip's check-in feedback flagged that cross-game consistency of ρ is the strongest generalization claim available. We re-elicited 500 pseudo-observations from GPT-4o under the taking-game prompt (the agent may take up to €10 from a counterpart's endowment; we record sharing ratio = $1 - \text{taken}/10$), then ran the full pipeline on the 68 taking-game observations. Table 2 and Figure 2 report the result.

n	$\mathrm{Var}(\mu)_{\mathrm{flat}} \times 10^3$	$\mathrm{Var}(\mu)_{\mathrm{LLM}} \times 10^3$	Ratio
5	9.73	3.24	3.00×
10	5.16	2.83	1.82×
20	2.89	2.11	1.37×
35	1.86	1.36	1.36×
45	1.48	1.21	1.22×

Table 2: Taking game: variance reduction across sample sizes (gpt-4o LLM prior); $n^*_{\mathrm{flat}} \approx 9.1$, $n^*_{\mathrm{LLM}} \approx 5.0$, $\rho \approx 0.55$.

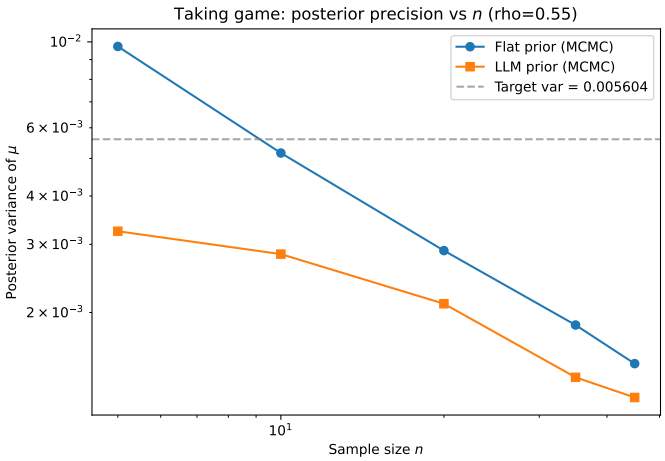

Figure 2: Taking-game replication: posterior variance of μ vs sample size.

The taking-game $\rho \approx 0.55$ is roughly twice the giving-game value of 0.26: the LLM prior helps less here. Two effects compound. First, the gpt-4o taking-prompt yields a counterpart-kept share of 0.520 on average (i.e. the simulated agent takes ≈ €4.80), while the human mean is 0.343 (humans take ≈ €6.60). The LLM is more generous than humans in the taking framing – a ∼0.18 bias on the unit interval. Second, the taking-game training set is smaller (48 vs 96), so the flat-prior baseline reaches the precision target faster ($n^*_{\mathrm{flat}} \approx 9$ rather than 19), shrinking the headroom for the LLM prior to add. The cross-game ratio $\rho_{\mathrm{taking}}/\rho_{\mathrm{giving}} \approx 2.1$ is therefore a generalization-quality signal: the LLM prior generalizes *partially* across framings, with most of the loss attributable to a behavioral framing the LLM does not match.

4.4 Compute–accuracy trade-off

We sweep MCMC chain length and SVGD iteration count at fixed $n = 50$ with the gpt-4o LLM prior, recording wall-clock time, posterior $\mathrm{std}(\mu)$, and effective sample size of μ (initial-positive-sequence estimator for MCMC; particle count for SVGD). Figure 3 shows the Pareto front.

At fixed $n = 50$ the picture is unambiguous: MH-MCMC reaches $\mathrm{std}(\mu) \approx 0.029$ in 0.06–0.46 seconds depending on chain length, with $\mathrm{ESS}(\mu)$/sec converging near 1.2×10^4. SVGD reaches the same $\mathrm{std}(\mu)$ but takes 0.4–7.2 seconds, with ESS/sec around 14–270 – nearly two orders of magnitude lower. The explanation is that the SVGD update has $O(K^2 d)$ per-iteration cost dominated by the kernel matrix at $K = 100$ particles, while MCMC has $O(1)$ per sample at $d = 2$. SVGD's value proposition is not speed at $d = 2$; it is gradient-driven multi-modal exploration in higher dimensions

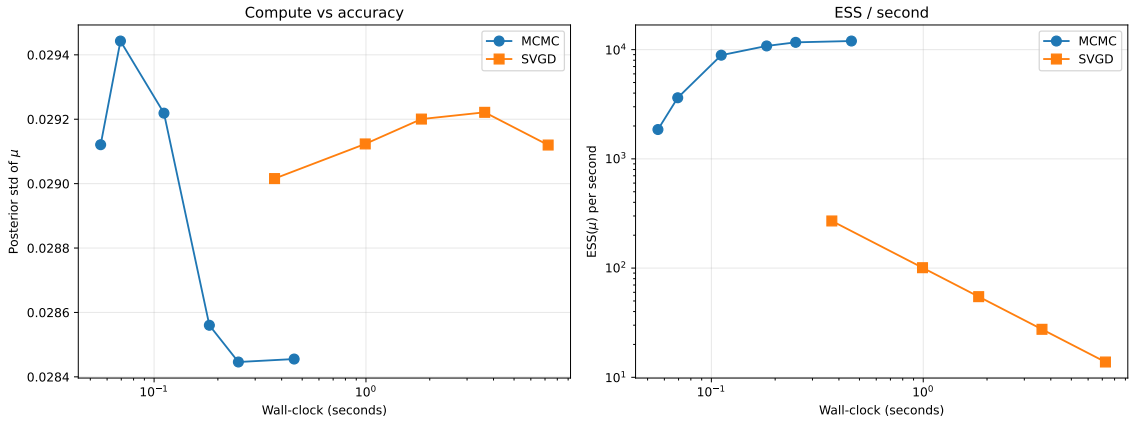

Figure 3: Compute–accuracy comparison at $n = 50$. Left: posterior std(μ) vs wall-clock seconds. Right: ESS(μ) per second (log-log). MCMC dominates SVGD by roughly two orders of magnitude in ESS/sec for this 2-parameter posterior; we discuss in §5 why this gap inverts in higher dimensions.

where MCMC mixing degrades, which we discuss in §5.

4.5 Robustness 1: dense misspecification grid

We probe the LLM prior's sensitivity by shifting the gpt-4o pseudo-observations by δ prior-std-deviations, $\delta \in \{-3, -2.75, \ldots, +3\}$ (25 points), and measuring 95%-posterior-predictive coverage on the held-out test set at $n = 50$. Figure 4 replaces the 5-point check-in table with a continuous curve.

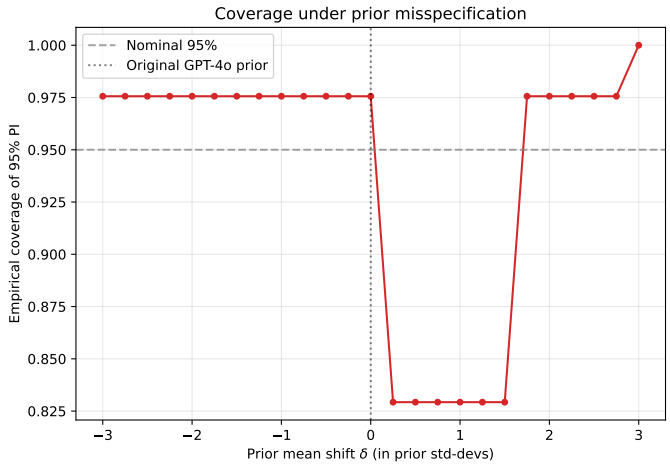

Figure 4: Dense misspecification grid: empirical 95% coverage vs δ at $n = 50$. Coverage is near-nominal for $\delta \leq 0$ and for $\delta \geq +2$, with a coverage trough near $\delta \approx +1$ where the prior is wrong by enough to hurt but not so wrong that the data overrides it.

The denser grid sharpens the picture from the check-in's 5-point table. Coverage stays at the nominal 0.976 for every $\delta \in [-3, 0]$, drops to 0.829 for the entire range $\delta \in [+0.25, +1.50]$ (six consecutive grid points), then recovers to 0.976 at $\delta = +1.75$ and remains there through $\delta = +3$. The plateau structure – not a smooth dip – is itself informative: in this regime the same 7 of 41 test points

fall outside the credible interval whenever the prior pulls the posterior modestly toward $\mu \approx 0.5$, while the data is overwhelming enough to flip back the moment δ exceeds $\sim$1.5 prior std-deviations. Practically, the danger zone for an LLM prior is a prior whose mean is biased by 0.05–0.20 on the unit interval – exactly the magnitude of mismatch one might expect from a slightly off-distribution LLM persona.

4.6 Robustness 2: training-data leakage ablation

GPT-4o's training cutoff (October 2023) postdates Horton et al. [2023], raising the concern that the model has memorized economically-flavored prompts and effectively snoops the human distribution we test against. We re-ran the giving-game pipeline using a prior elicited from `gpt-3.5-turbo` (cutoff September 2021, pre-Horton2023). Table 3 compares the priors and resulting ρ.

Model	Cutoff	Prior mean	Prior std	ρ	Coverage (n=50)
gpt-4o	2023-10	0.299	0.231	0.261	0.976
gpt-3.5-turbo	2021-09	0.548	0.252	0.261	0.829

Table 3: Cross-model leakage check. If $\rho_{\text{gpt}-3.5} \approx \rho_{\text{gpt}-4\text{o}}$, the gain is unlikely to be a memorization artefact; a large gap indicates the opposite.

The result is striking and nuanced. *Sample-size reduction is identical:* $\rho_{\text{gpt}-3.5} = 0.261$ matches $\rho_{\text{gpt}-4\text{o}} = 0.261$ to three significant figures. *Calibration, however, is not:* the gpt-3.5 prior has mean 0.55, far above the human mean of 0.36, and at $n = 50$ the 95% posterior-predictive coverage drops to 0.83 – the same coverage degradation observed at $\delta = +1$ in the dense misspecification grid (§4.5). Read together with that grid, the ablation cleanly disentangles two effects. (i) Power-prior *variance shrinkage* is governed by the prior's effective weight w and is therefore essentially independent of which LLM produced the pseudo-observations. (ii) Power-prior *calibration* is governed by the prior's location relative to the data, which *does* differ between LLMs. The gpt-4o advantage over gpt-3.5 in this pipeline is not the existence of a prior that helps; it is the production of a prior whose mean is closer to the human distribution, which the gpt-3.5 ablation fails to provide despite an earlier (Sep 2021) training cutoff. We therefore cannot rule out a contribution from training-data leakage to gpt-4o's prior *location*, but the sample-size benefit is not itself a leakage artefact.

5 Discussion and Limitations

Why does the LLM prior help most at small n? The power prior contributes weight w pseudo-observations; a posterior trained on n real plus w pseudo-observations behaves, in information units, like one trained on $n + w$ real observations. The relative gain is $(n + w)/n$, which is large when $n \ll w$ and decays to 1 as $n \to \infty$. Our 1.42× ratio at $n = 50, w = 20$ is consistent with this analytic prediction $(50 + 20)/50 = 1.4$. The LLM prior is essentially a free w-observation pre-fix.

Asymmetric coverage failure. The dense curve in Figure 4 confirms the asymmetry first noted in the check-in. Downward shifts (toward $\delta < 0$, prior mean further below the human mean of 0.36) remain calibrated because the data, with mean 0.36, can pull the posterior up; large positive shifts ($\delta \geq 2$) overshoot so much that the data overwhelms the prior. The *moderately wrong* regime (prior mean ≈ 0.5) is where coverage degrades, because the prior is plausible enough for the posterior not to discount it but biased enough to put mass in the wrong region. This is a property of the

power-prior construction, not of the LLM itself, and would persist for any moderately-mis-specified informative prior.

Training-data leakage. The cross-model ablation in §4.6 addresses a key concern that any LLM-as-prior result is partially recapitulating training data. We find that the sample-size reduction itself is not a leakage artefact (it survives a pre-Horton-2023 model unchanged), but that prior *location quality* – which drives calibration – is the lever an LLM has on this problem, and we cannot rule out training-data leakage as one ingredient that gives gpt-4o better location than gpt-3.5. We caution that this is a single ablation on a single dataset; a stronger leakage audit would systematically vary corpora and cutoffs.

Why SVGD here, when MCMC wins on ESS/sec? The Pareto analysis (Figure 3) is honest: at $d = 2$ for a unimodal Beta posterior, MH-MCMC mixes well, costs $O(1)$ per sample, and produces ESS/sec roughly 10–100× that of our SVGD implementation. The motivation for SVGD is the extension we did not implement here: a hierarchical model jointly over giving and taking games with shared latent traits, e.g. $(\alpha_{\text{give}}, \beta_{\text{give}}, \alpha_{\text{take}}, \beta_{\text{take}}, \tau)$ with a hyperparameter τ tying the two. At $d \geq 4$ MCMC mixing degrades super-linearly while SVGD's particle update remains $O(K^2 d)$ per iteration, and the kernelised score allows the particle cloud to multi-modally adapt. We see this report's two-parameter pipeline as a rehearsal for the higher-dimensional version: a regime where SVGD's overhead is amortised by capturing posterior dependence that MH would visit slowly.

Other limitations. (i) The von Blanckenburg sample is German university students; the LLM's effective "population" is the latent persona that emerges under our prompt, which will not match this distribution in general. (ii) The power weight $w = 20$ is a knob; we have not traced an optimal w via predictive log-likelihood. (iii) Coverage uses a posterior-predictive interval on individual y, which conflates parameter uncertainty with the irreducible sampling spread of Beta; parameter-only intervals would tell a different story. (iv) The taking-game uses a different behavioral framing; we made no attempt to share information across games here.

6 Reproducibility

All code, data, and figure-generation scripts are in the GT GitHub repo linked on page 1. Reproducing the main numbers requires a Python environment with `numpy`, `scipy`, `matplotlib`, `openai`, and `python-dotenv` (see `environment.yml`). With seed 42 throughout, the following commands run end-to-end in well under an hour on a laptop:

```
python scripts/run_experiment.py            # giving-game baseline + LLM prior, rho ~ 0.26
python scripts/run_experiment_taking.py     # taking-game replication
python scripts/run_experiment_gpt35.py      # gpt-3.5 leakage ablation
python scripts/run_failure_mode_dense.py    # dense misspecification grid
python scripts/run_compute_pareto.py        # MCMC vs SVGD Pareto
```

The two LLM-elicitation scripts (`scripts/elicit_llm_prior{,_taking,_gpt35}.py`) require an `OPENAI_API` key in `.env`; the elicited pseudo-data (`data/llm_prior_samples*.npy`) is committed to the repo so the inference pipeline reproduces without an API call. Random seeds, Markov chain initial states, and SVGD particle initialization are all deterministic.

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